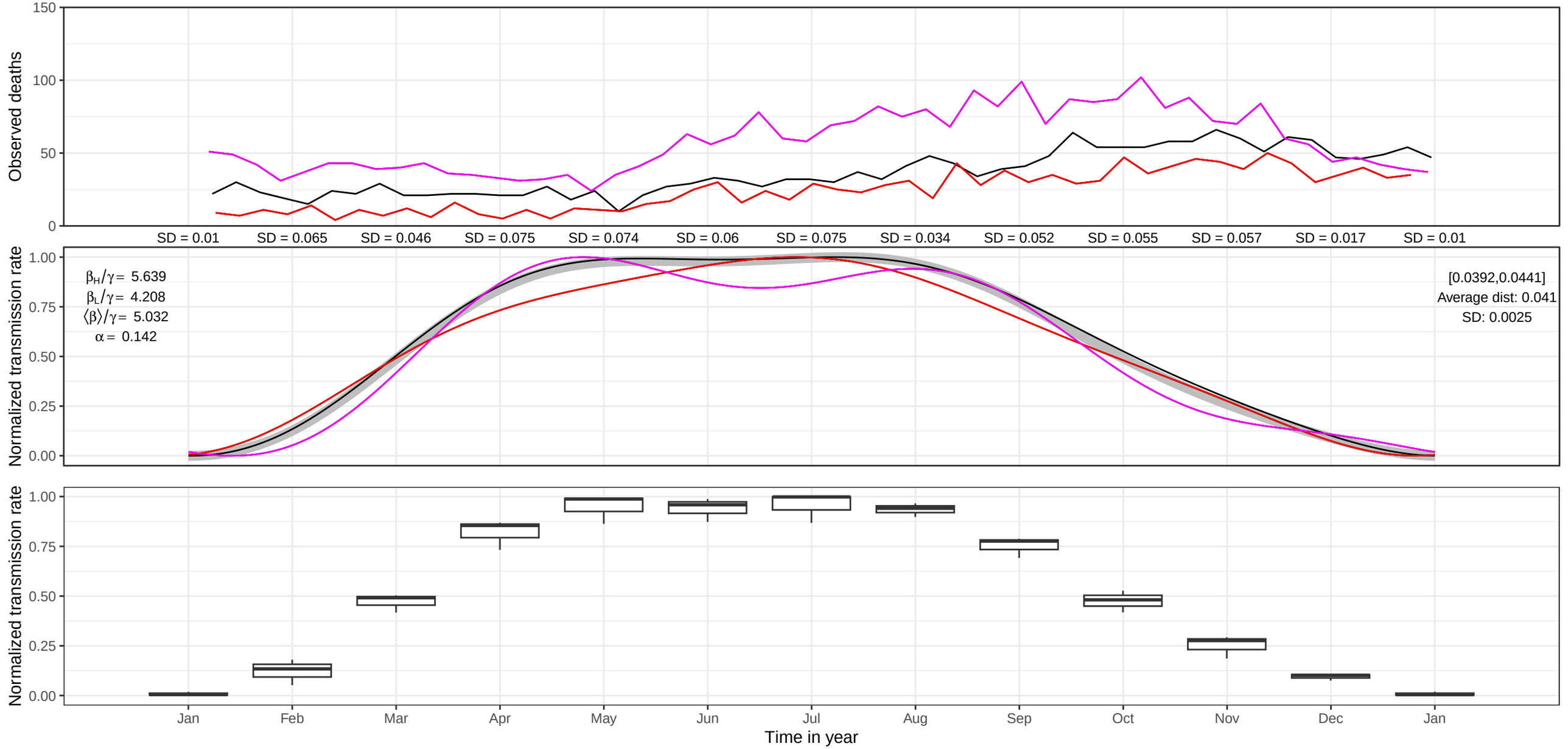
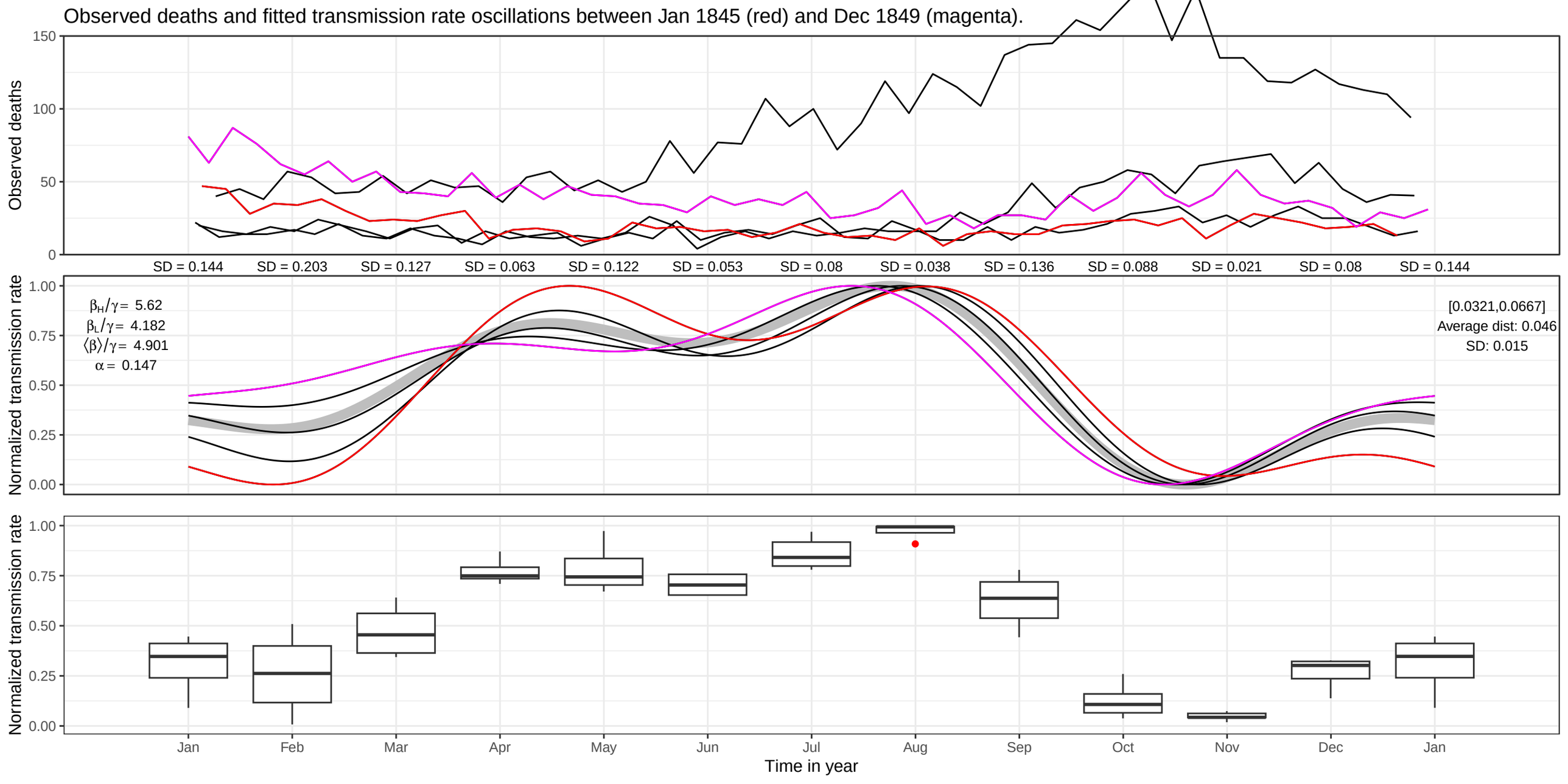
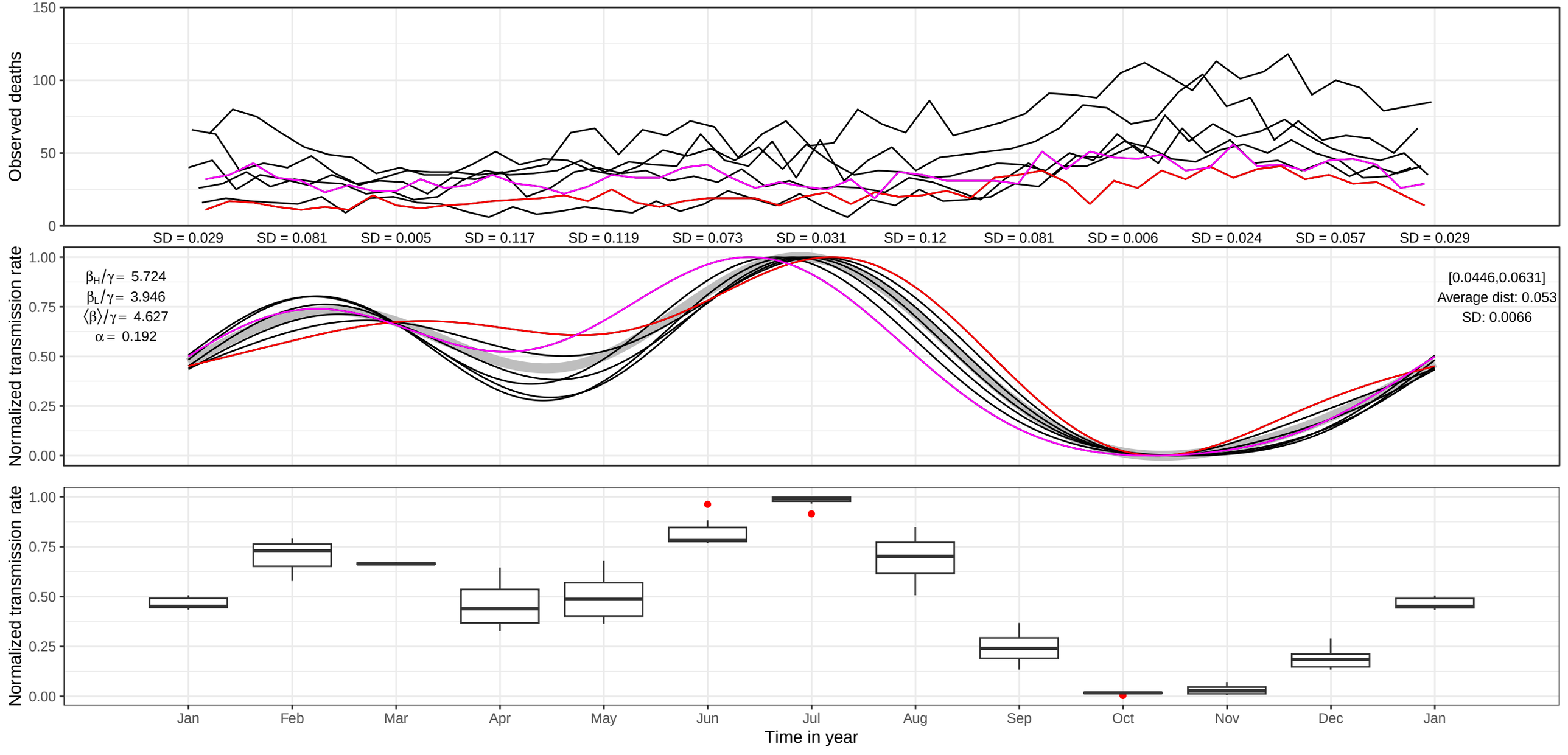
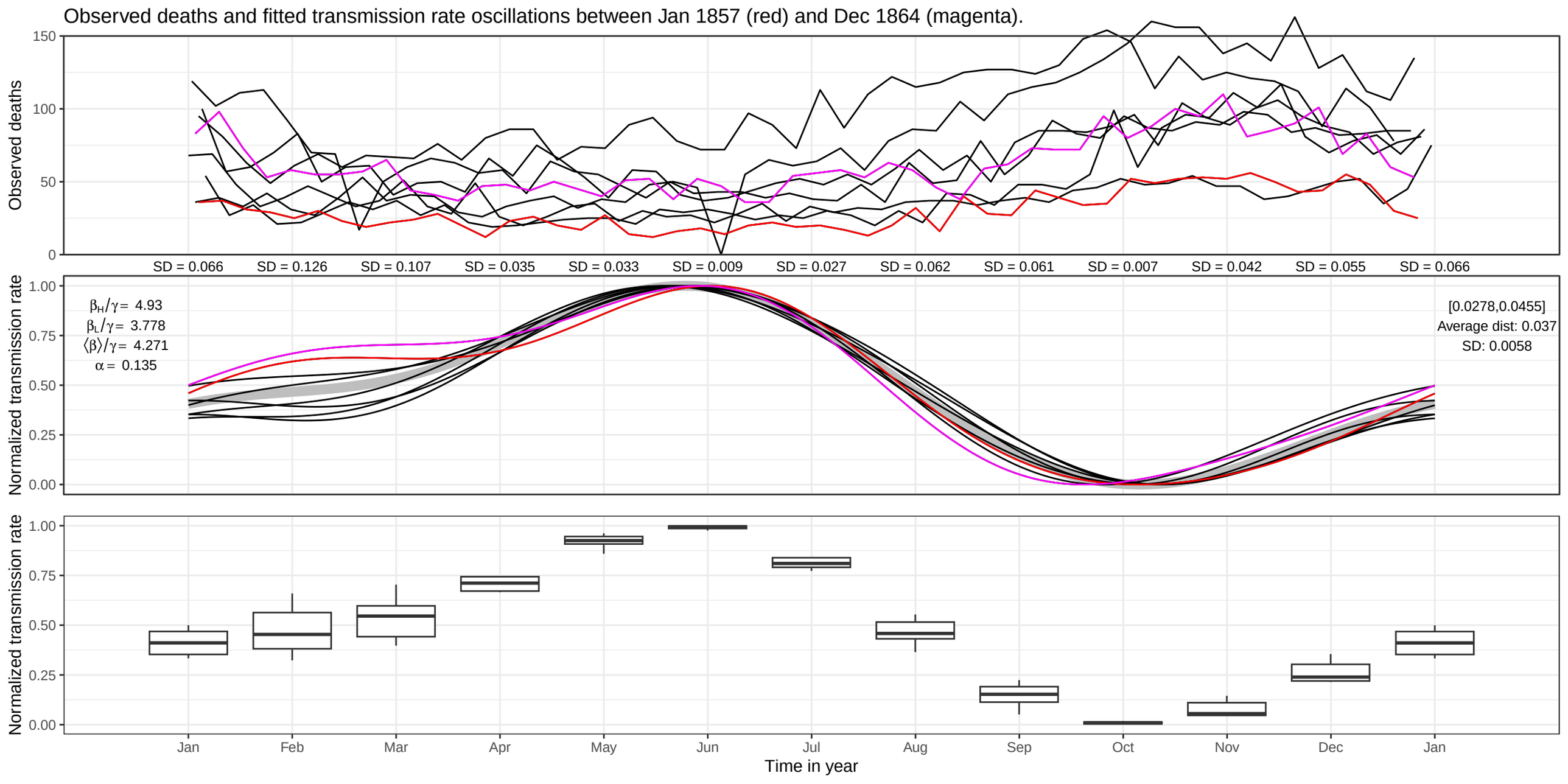
Observed deaths and fitted transmission rate oscillations between Jan 1842 (red) and Dec 1844 (magenta).

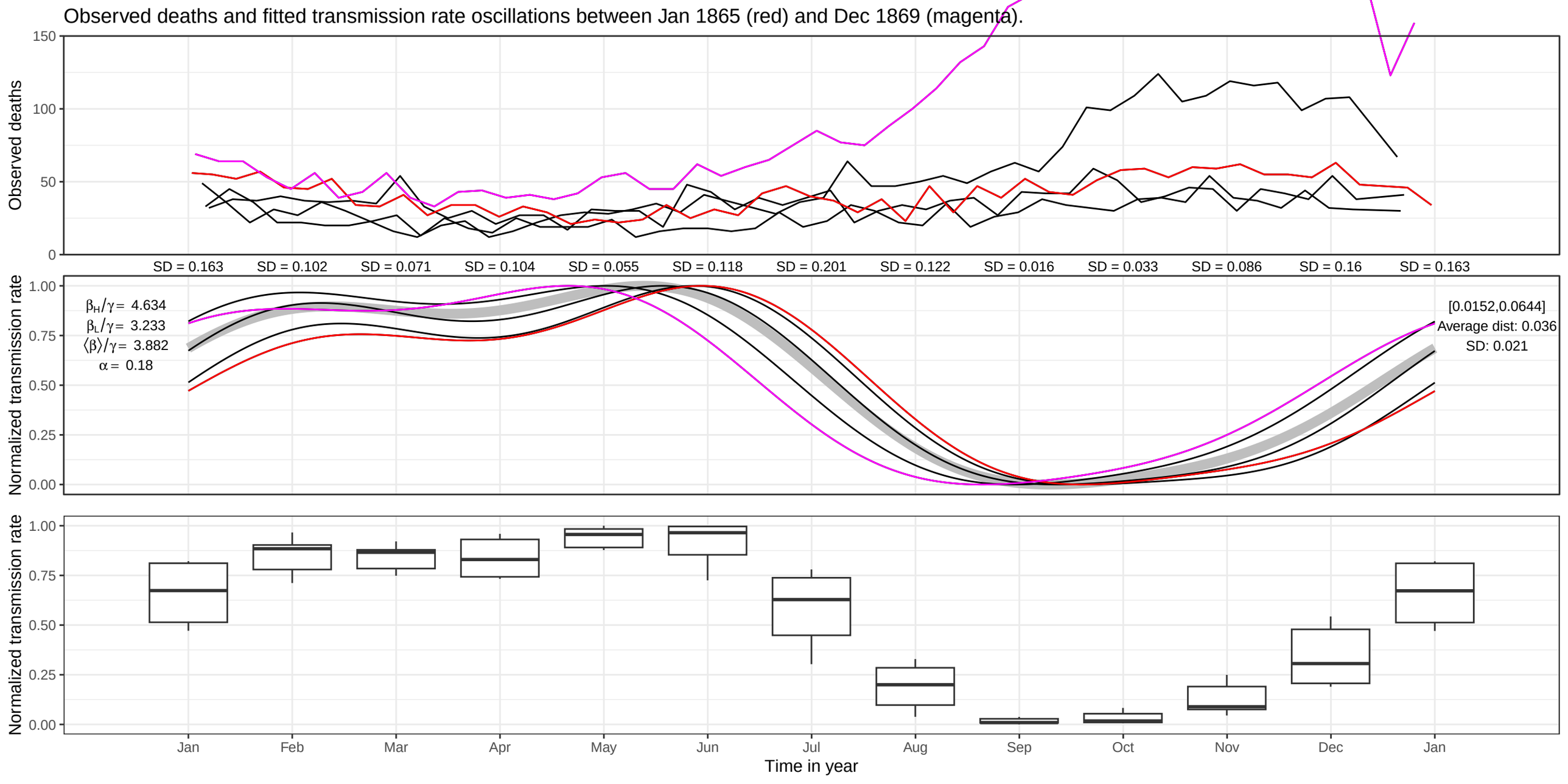


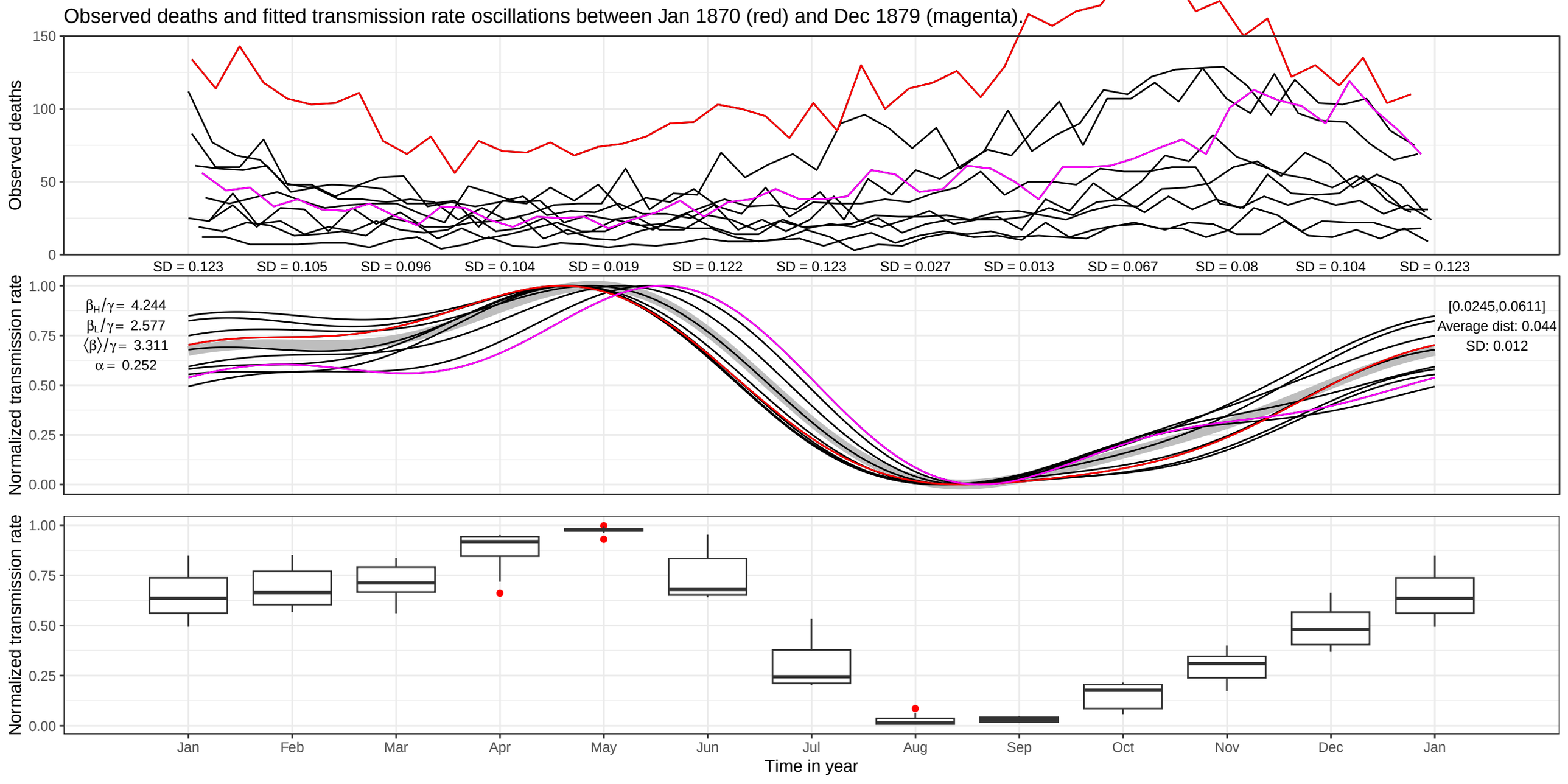

Observed deaths and fitted transmission rate oscillations between Jan 1845 (red) and Dec 1849 (magenta).
Observed deaths
150
100
50
0
SD = 0.144
SD = 0.203
SD = 0.127
SD = 0.063
SD = 0.122
SD = 0.053
SD = 0.08
SD = 0.038
SD = 0.136
SD = 0.088
SD = 0.021
SD = 0.08
SD = 0.144
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ= 5.62
βL/γ= 4.182
⟨β⟩/γ= 4.901
α= 0.147
[0.0321,0.0667]
Average dist: 0.046
SD: 0.015
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

Observed deaths and fitted transmission rate oscillations between Jan 1850 (red) and Dec 1856 (magenta).

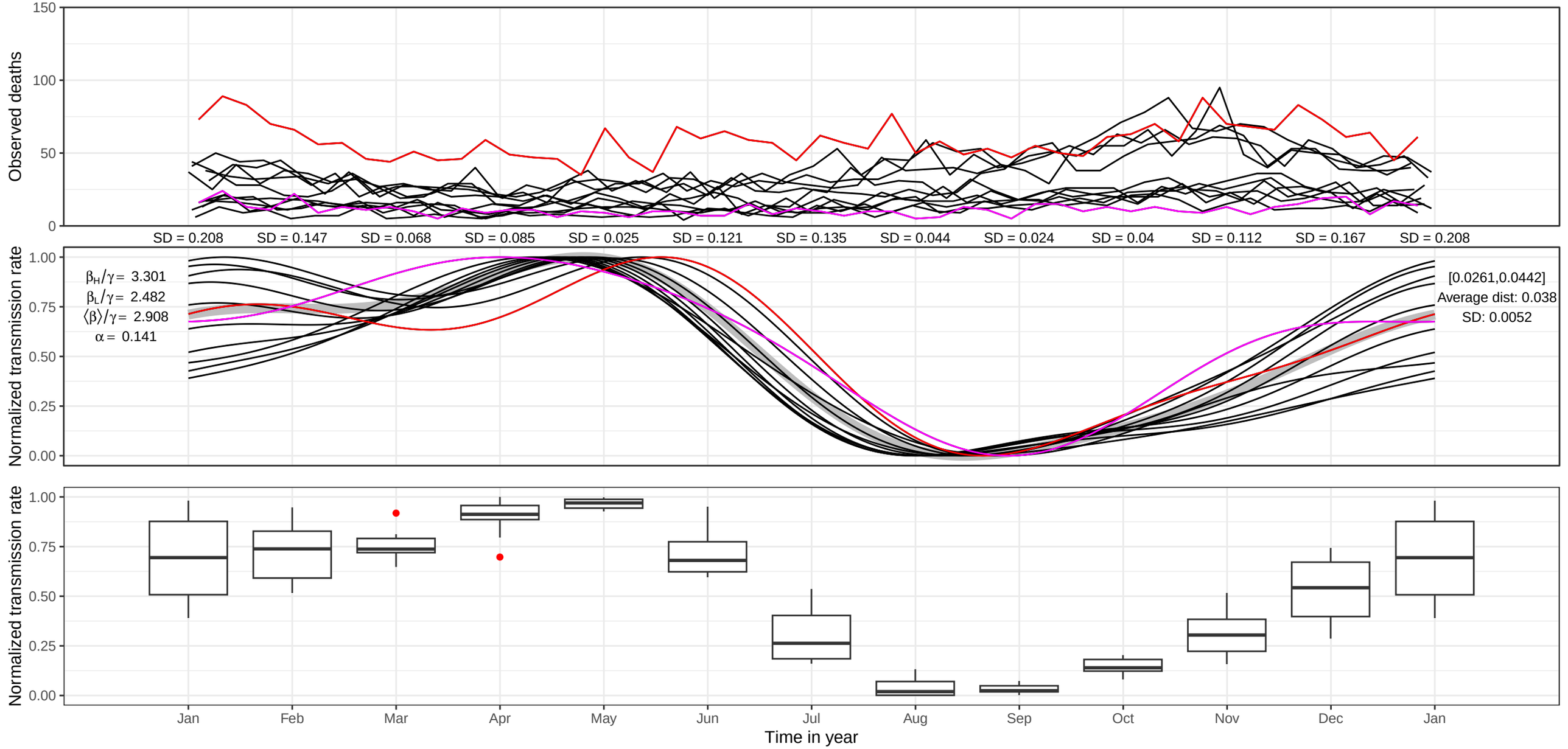



Observed deaths and fitted transmission rate oscillations between Jan 1857 (red) and Dec 1864 (magenta).
Observed deaths
150
100
50
0
SD = 0.066
SD = 0.126
SD = 0.107
SD = 0.035
SD = 0.033
SD = 0.009
SD = 0.027
SD = 0.062
SD = 0.061
SD = 0.007
SD = 0.042
SD = 0.055
SD = 0.066
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ= 4.93
βL/γ= 3.778
⟨β⟩/γ= 4.271
α= 0.135
[0.0278,0.0455]
Average dist: 0.037
SD: 0.0058
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

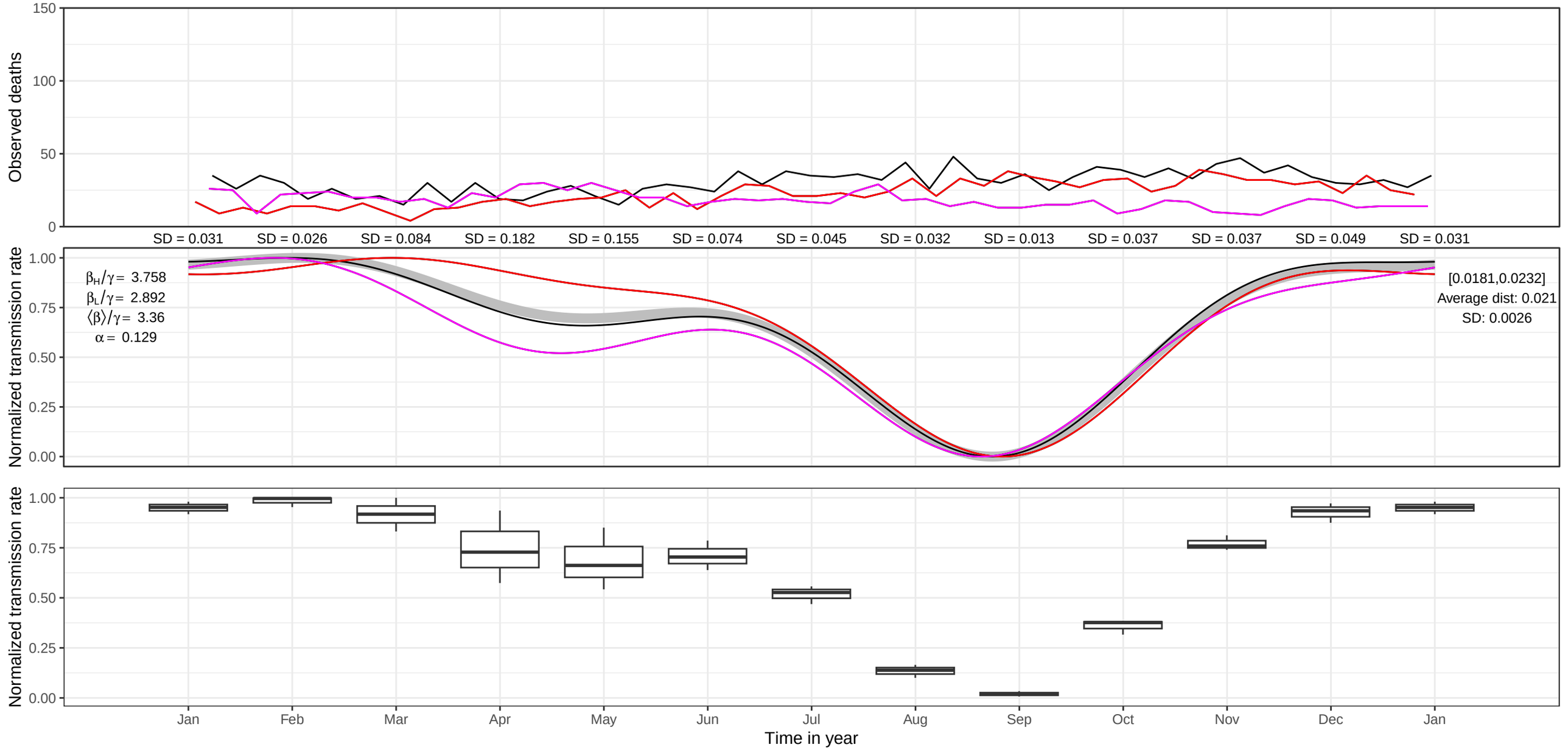

Observed deaths and fitted transmission rate oscillations between Jan 1865 (red) and Dec 1869 (magenta).
Observed deaths
150
100
50
0
SD = 0.163
SD = 0.102
SD = 0.071
SD = 0.104
SD = 0.055
SD = 0.118
SD = 0.201
SD = 0.122
SD = 0.016
SD = 0.033
SD = 0.086
SD = 0.16
SD = 0.163
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ = 4.634
βL/γ = 3.233
⟨β⟩/γ = 3.882
α = 0.18
[0.0152,0.0644]
Average dist: 0.036
SD: 0.021
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year


Observed deaths and fitted transmission rate oscillations between Jan 1870 (red) and Dec 1879 (magenta).
Observed deaths
150
100
50
0
SD = 0.123
SD = 0.105
SD = 0.096
SD = 0.104
SD = 0.019
SD = 0.122
SD = 0.123
SD = 0.027
SD = 0.013
SD = 0.067
SD = 0.08
SD = 0.104
SD = 0.123
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
$\beta_H/\gamma =$ 4.244
$\beta_L/\gamma =$ 2.577
$\langle\beta\rangle/\gamma =$ 3.311
$\alpha =$ 0.252
[0.0245,0.0611]
Average dist: 0.044
SD: 0.012
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

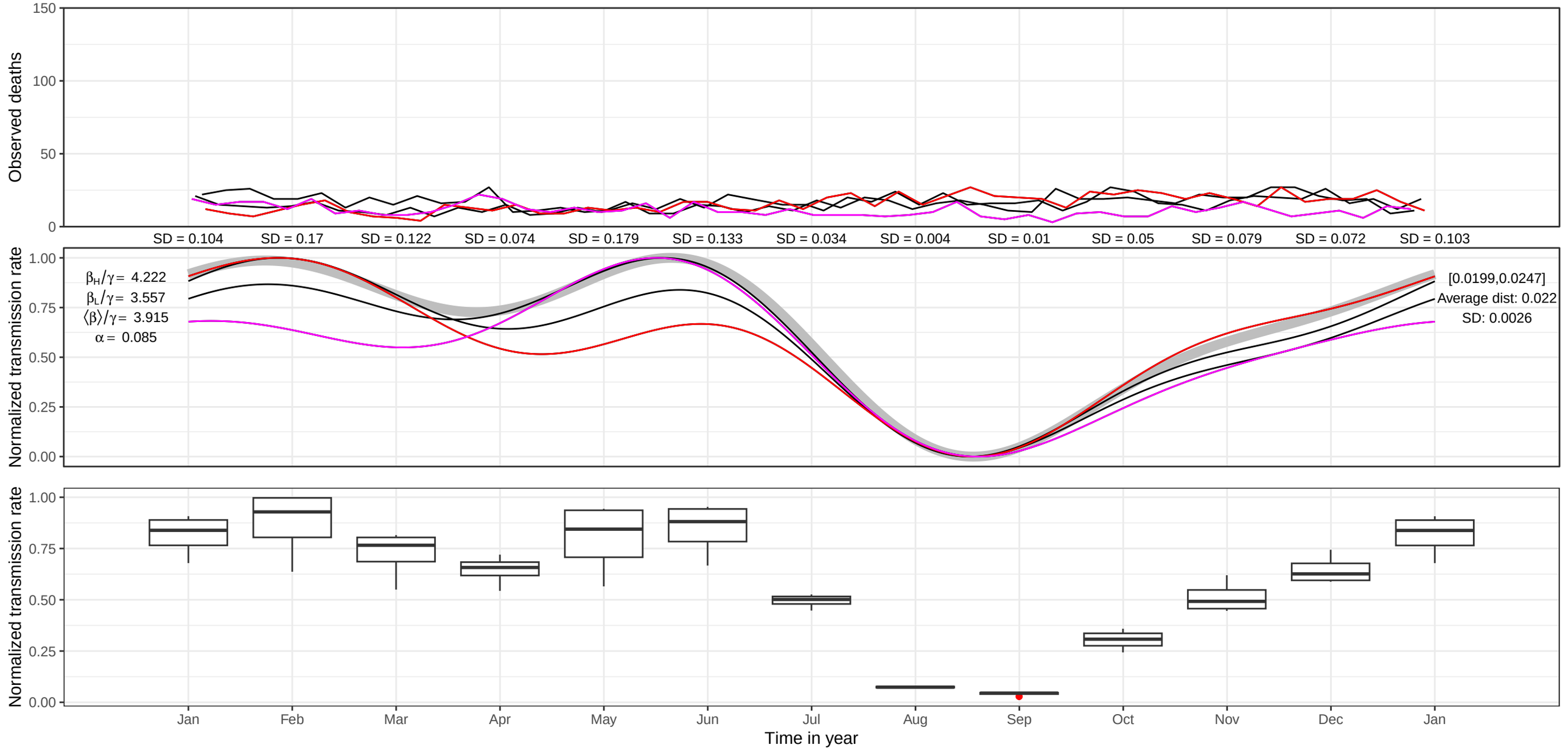
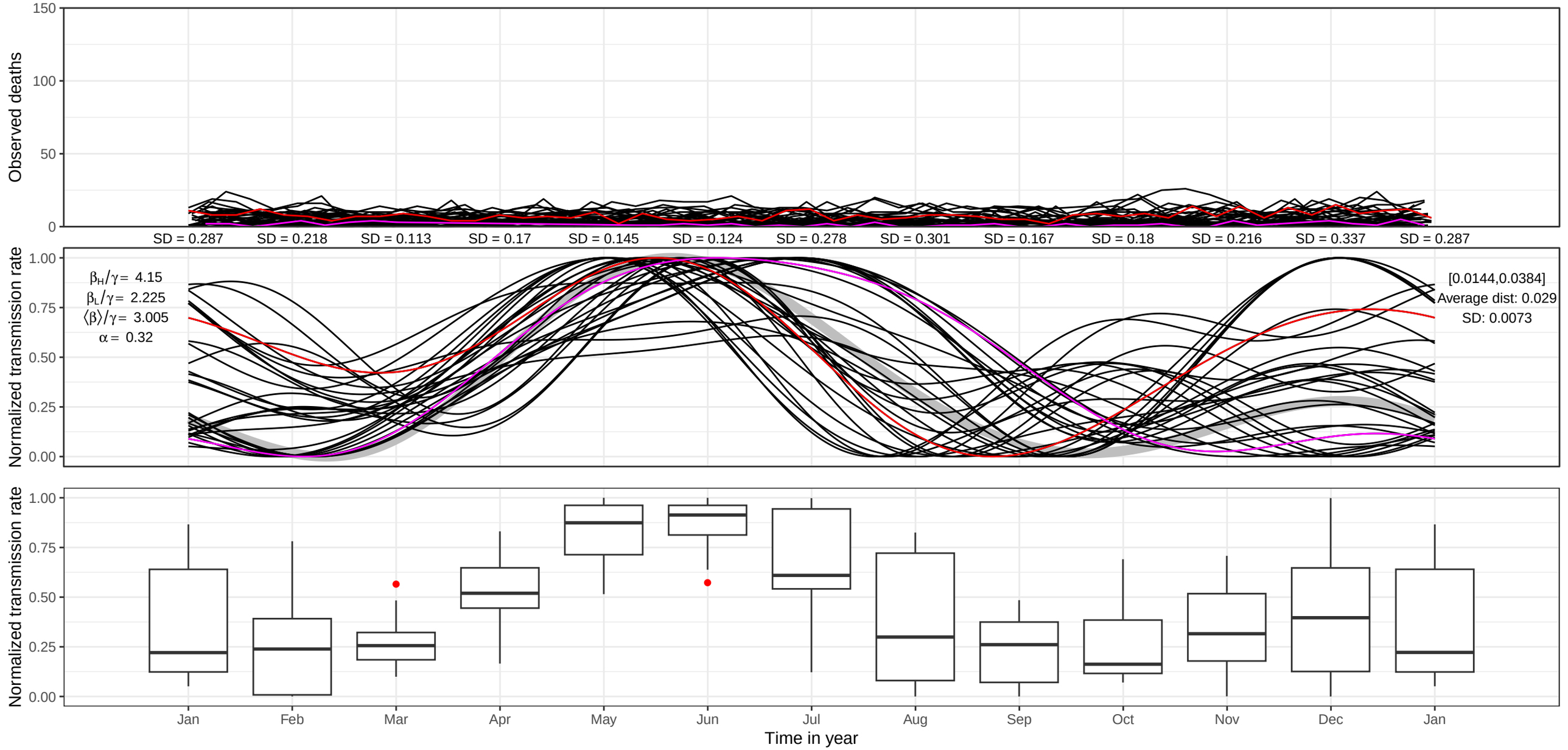
Observed deaths and fitted transmission rate oscillations between Jan 1880 (red) and Dec 1891 (magenta).



Observed deaths and fitted transmission rate oscillations between Jan 1892 (red) and Dec 1894 (magenta).



Observed deaths and fitted transmission rate oscillations between Jan 1895 (red) and Dec 1898 (magenta).



Observed deaths and fitted transmission rate oscillations between Jan 1899 (red) and Dec 1929 (magenta).